

can be considered a sub-set of AI. Toyota has brought together AI, ML and Big Data for the generation of highly responsive autonomous systems, which can help people who cannot drive properly.

Deep learning

ML is implemented using DL. Many activities in AI are carried out using DL. Complex tasks are broken down into smaller ones, which then become manageable. Software in DL learns activities and mimics these in the brain's neuron layers.

Today, DL techniques play a vital role in autonomous driving and advanced driver assistance systems—what goes on inside the vehicle. DL plays a major role in the workings of the outside of the vehicle, and during service, sales and manufacturing phases.

Continuous improvements are being made in the service phase using DL. A partnership has been formed between Bosch and NVIDIA to enhance the features and quality of autonomous vehicles. As part of this partnership, Bosch will develop a supercomputer that works in autonomous vehicles and DL will offer technology to power it. A new level of AI in cars such as Mercedes Benz and Audi is expected in the future, thanks to Bosch and NVIDIA.

The Internet of Things

In the automobile industry, the Internet of Things (IoT) has made an indelible mark. It is poised to become even more powerful in the years to come. Vehicles now come with a host of features, like embedded connectivity applications, geo-analytic capabilities using Big Data and smart sensors.

Vehicle owners now enjoy many attributes, as given below.

- A manufacturer or dealer can share firmware updates of a vehicle via over-the-air software.
- A vehicle's performance data can be sent to the service centre, dealer and manufacturer if it is scheduled for repair or service on a specific day.
- Organisations with a large number of vehicles can handle the same using enhanced safety parameters.
- Through IoT processes, improved quality in manufacturing is possible.
- Smart sensors in a car can alert

relevant medical personnel in case of an emergency.

Cognitive capabilities

Cognitive analysis imitates the behaviour of the brain by looking at its data mining capabilities and behaviour patterns. This revolutionises not only small cars with AI capabilities but also large, 18-wheel carrier trucks.

Cognitive systems need to have a much deeper understanding of unstructured data, as these must work like a human would in a real-life situation. To decide on how to respond naturally in real time, many insights are drawn from large amounts of unstructured data.

Dynamic operating conditions can also be handled by cognitive capabilities. These capabilities are already being incorporated into vehicles by many car manufacturers. For example, to add cognitive capabilities to its cars, BMW has entered into a partnership with IBM.

Infotainment systems

AI transforms in-vehicle infotainment systems in a significant manner. Demand for high-quality software and hardware solutions has soared due to this, as these have to be compatible with AI.

We can expect a spike in features such as natural languages database, gesture recognition, driving pattern monitoring, eye tracking and speech recognition. This will eventually go a step further to detection units based on radars, sensor fusion engine control units, camera-attached machine vision systems and evaluation of driver condition.

Machine-human interfaces attached to vehicles can monitor and act in accordance with algorithms collected from neural networks based on cloud. These are utilised to perform a number of advanced tasks.

Conclusion

We will soon witness the cloud-to-air phenomenon (a new kind of car driving) through AI capabilities. Developers have created applications that have taken AI to a new level of excellence, thanks to great computational power.

AI is not only about self-driving cars. It also helps in developing cars that can perceive the environment and navigate through hindrances that may come up while driving. **EFY**



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